2. If the injector passes these tests, it is satisfactory for further use. However, failure to pass any one of the tests indicates that defective or dirty parts have been assembled. In this case, the injector must be disassembled again, cleaned, inspected, reassembled, and tested again.

3. If an injector is not to be used immediately, shipping caps should be installed on the injector caps. The injector test oil remaining in the injector after the fuel output test will serve as a rust preventive while the injector is in storage.

INJECTOR INSTALLATION

Before installing a new or reconditioned injector in an engine, the carbon deposits must be removed from the beveled seat of the injector hole tube in the cylinder head. If this carbon is not removed, the injector may be cocked during installation resulting in undue stresses being exerted against the spray tip. This could ultimately result in a fractured spray tip.

Injector tube bevel seat reamer should be used to clean the carbon out of the tube before installing the injector. Refer to "Reaming" in "CYLINDER HEAD" section of this manual. Care must be exercised to remove ONLY the carbon so that the proper clearance between the injector body and cylinder head is maintained. The flutes of the reamer should be packed with heavy grease to retain the carbon removed from the tube.

1. Refer to figure 7 and insert the injector into the injector tube with the locating dowel registering with the locating hole in the cylinder head and with the pin on the injector control tube lever registering with the injector control rack lever.

2. Place injector clamp over stud. Install special washer and nut and tighten nut to torque of 20 to 25 ft.-lbs. Check to make sure clamp does not interfere with valve or injector springs.

NOTE: Check injector rack for free movement. Excess torque will cause the injector control rack to stick or bind.

3. Move rocker arm assembly into position and secure rocker arm bracket bolts to torque of 90 to 100 ft.-lbs. (fig. 8).

4. Connect fuel pipes (fig. 8) to injector and fuel connectors.

5. As a precautionary measure against any possibility of scoring injector parts upon initial installation due to lack of lubrication, any entrapped air should be bled from the injector before tightening the connections on the fuel outlet side of the injector. This may be accomplished by cranking the engine briefly with the injector rack in the NO FUEL position and then tightening the fuel pipe connection.

6. When installing injector fuel lines, connections should be tightened only enough to prevent leakage of fuel. Excess tightening may result in the flared end of the fuel line being twisted or fractured. When all injector fuel lines are installed, the engine should be run long enough (during tune-up procedure) to check for leaks. Should leaks occur, the connections should be tightened only enough to stop the leak.

7. Following installation of the injectors in the engine, a complete engine tune-up should be performed. However, if only one injector has been removed and replaced, and other injectors and the governor adjustment have not been disturbed, it will only be necessary to adjust the clearance of the valves, valve bridge, and time the injector for the one cylinder, and position the control rack on the cylinder involved to correspond with that of the other racks.

CONTROL RACK POSITIONING

1. Whenever injector control tube has been removed, care should be exercised when replacing assembly to be sure that no bind exists between tube levers and injector racks to hamper free operation of the latter. The levers are free to move endwise on tube a limited distance.

2. Whenever the adjusting screw and the levers are tightened, levers should be so positioned on tube that operating lug at lower end of lever does not bind in slot of injector rack. Furthermore, the brackets holding tube assembly to cylinder head should be so positioned that coil spring on control tube will return injector racks to **NO FUEL** position freely.

3. Control tube and bracket assembly may be shifted by loosening bracket-to-cylinder head bolts and tapping brackets lightly to change their positions on cylinder head, then retightening hold-down bolts. Instances may arise when an injector rack does not move freely in the injector or the plunger in the bushing. This is particularly true if injector has been used over a long period of time without cleaning.

4. If no bind exists between rack control tube and injectors, and the tension spring on control tube fails to return racks freely to **NO FUEL** position when control tube is disconnected from governor fuel rod, the cause may be due to a sticking injector rack. A sticking injector rack may be located by checking each injector as follows: